Shelden Rattray

(470) 429-4524 | shelden.rattray1@gmail.com | linkedin.com/in/shelden-rattray | github.com/sheldenr | sheldenr.dev

Education

Georgia State University (Honors College) *Computer Science, B.S.* *Expected May 2027*

Awards $2500 Delta Analytics Scholarship Recipient, 5x President's List Recipient **GPA: 4.00/4.00**

Experience: STEM Tutor for undergraduate Computer Science Courses.

Experience

Cloud Platform Engineering Intern | Fiserv (Fortune 500) May 2026 – Present

- Architected and modularized infrastructure-as-code (IaC) using **Terraform** to automate the provisioning of highly available multi-region architectures across **AWS** and **Azure**.
- Orchestrated containerized workloads using **Kubernetes (K8s)** and Docker, configuring auto-scaling policies and resource limits to optimize cluster efficiency and ensure zero-downtime deployments.
- Designed and implemented high-performance caching strategies using **Redis Cache**, drastically reducing database read latencies and enhancing application throughput under peak traffic loads.
- Engineered robust **CI/CD pipelines** paired with custom **Bash/Python scripts** to automate system maintenance, VM patch management, and configuration compliance across hybrid cloud environments.

Software Engineering Intern | CuesHub Jul 2025 – Aug 2025

- Built 10+ interfaces in for CuesHub mobile application for real-time heart rate and accelerometer visualizations.
- Implemented in-app logging utilities and structured exception handling to improve issue traceability and application stability.
- Optimized state management and caching mechanisms for 30% improvement in rendering performance.

Projects

Distributed Ledger & Transaction Engine | *Go, PostgreSQL, Redis, Docker, Terraform* – GitHub

- Architected a concurrent double-entry bookkeeping engine in **Go**, using deterministic row-level pessimistic locking and **Redis** distributed mutexes to guarantee zero double-spends and eliminate deadlock risks.
- Implemented two-tier payload-hashed idempotency caching across **Redis** and **PostgreSQL** for sub-millisecond safe request replays, automated via **Terraform** for multi-tier **AWS** deployments.

Polysia | Full-Stack Web Application | *React, Firebase, PostgreSQL, RAG* – Website – GitHub

- Built and deployed a full-stack web application that tracks, creates and reads stories in Mandarin Chinese to assist language learners.
- Implemented a simple model that takes user character retention data via a simple SRS-based learning system and maps the data into a custom Adaptive Learning Model, accurately determining each user's learning rate.
- Integrated **LLM-based** conversational feedback pipelines powered by DeepSeek and Azure Neural TTS.

Rhythm Game Performance Calculator | Browser Extension | *JavaScript* – GitHub

- Developed a browser extension with **10,000+** installs, featuring a custom API integration layer for fetching, normalizing, and processing game data at scale.
- Optimized asynchronous workflows and caching to reduce load times and ensure reliable data accuracy.

Technical Skills

Languages: Go, Python, Java, JavaScript, TypeScript, C#, Dart, SQL, Bash, HTML/CSS
Cloud & DevOps: AWS, Azure (AZ-900), GCP, Kubernetes, Terraform, Docker, CI/CD, Git, GitHub
Frameworks & Tools: React, Spring Boot, Redis, Firebase, PostgreSQL, TensorFlow, PyTorch, NumPy, Pandas

Leadership & Other Experience

Hackathon | 1st Place | Truist & Deloitte CTF Hackathon

- Captured 10/10 flags across real-world cybersecurity challenges on the HackTheBox's Capture The Flag platform.

Colorstack at GSU | Technical Director

- Architected and deployed the core web applications and underlying infrastructure supporting our digital interfaces.
- Manage and mentor 2 software engineering interns, conducting code reviews and guiding feature delivery across deployments.

CS Club @ GSU | Professional Development Director

- Lead career readiness initiatives, organizing technical interview prep sessions, resume reviews, and industry speaker workshops for club members.